

Steering Emergent Morphogenesis: A Real-Time Control Interface and Differentiable Digital Twin for Synthetic Protocells

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Abstract

Continuous cellular automata and reaction-diffusion fields can simulate complex biological self-organization, yet translating these models into physical wet-lab protocells reveals a major control bottleneck: non-linear dynamical boundaries cannot be anticipated through static threshold monitoring, causing operators to inadvertently drive non-equilibrium vesicles into irreversible failure states such as osmotic lysis or starvation. Here, we present *cgen*, an open-source real-time control interface that couples the differentiable SpudLenia protocell model to benchtop microfluidics through an online predictive stability filter. By unrolling forward simulation rollouts ($T_{\text{pred}} = 120$ steps; 14.4 s physical lookahead) in 141.6 ms on consumer GPU hardware, the system estimates local finite-time Lyapunov exponents to detect impending dynamical instabilities well before physical fluid transport delays (5–30 s) take effect. Across 500 simulated perturbation trials, predictive Lyapunov filtering provides an 8.4 s mean early-warning lead time, reducing osmotic rupture rates to 7.6% compared to 58.4% under standard threshold monitoring. We interface this predictive engine with continuous hardware controls and an open-source biomaker hardware pipeline utilizing laser-cut acrylic giant unilamellar vesicle (GUV) trapping arrays and cell-free transcription-translation (TX-TL). The complete interface, simulation engine, and microfluidic hardware files are released open-source.

Introduction

A core ambition of artificial life is synthesizing autonomous, self-replicating physical entities from non-living matter (Rasmussen et al., 2004; Adamala et al., 2026). In computational domains, continuous cellular automata such as Lenia (Chan, 2019, 2020) and Neural Cellular Automata (Mordvintsev et al., 2020; Niklasson et al., 2021) generate rich repertoires of decentralized morphogenesis. Recently, the SpudLenia framework (Oduniyi et al., 2026) demonstrated that cytokinesis (binary cell division) can emerge autonomously in continuous fields without complex active motor proteins, driven instead by membrane-associated steric protein crowding and localized growth inhibition.

However, translating continuous *in silico* morphogenetic models into living or synthetic wet-lab systems exposes a severe control challenge. In the physical wet lab, protocells—such as Giant Unilamellar Vesicles (GUVs)—are

open, dissipative thermodynamic structures operating far from equilibrium. Driving a protocell through cytokinesis requires carefully balancing lipid nutrient feeding, protein expression, and furrow constriction over 10 to 20 minutes. Standard laboratory interfaces rely on static post-hoc monitoring dashboards and numerical text entry that provide visual readouts but lack predictive stability feedback. Because physical microfluidic perfusion lines impose fluid transit delays of 5 to 30 seconds, an operator relying on instantaneous microscopy observations frequently drives the protocell across irreversible boundaries—either bursting the vesicle via osmotic overfeeding or collapsing it through starvation. A central insight of cybernetic control is that steering such open non-equilibrium systems requires continuous circular feedback loops between observation, predictive forward models, and actuation (Wiener, 1948; Ashby, 1956).

In this paper, we introduce *cgen*, a real-time control interface and predictive digital twin designed to steer continuous morphogenetic systems. Our core contributions are:

- Three-Level Abstraction Mapping:** We establish an explicit mapping from continuous cellular automata field parameters to physical microfluidic actuators (syringe flow, optical induction) and continuous interactive hardware controls.
- Online Predictive Stability Filter:** By unrolling truncated forward simulation trajectories ($T_{\text{pred}} = 120$ steps; 14.4 s physical lookahead) in 141.6 ms on consumer GPU hardware, we calculate finite-time local Lyapunov exponents $\lambda(t)$ to preempt dynamical bifurcations before physical fluids arrive.
- Empirical Stability Benchmark:** In 500 stochastic simulation trials, predictive Lyapunov filtering achieves an 8.4 s warning lead time, reducing osmotic rupture rates from 58.4% to 7.6%.
- Open Biomaker Wet-Lab Pipeline:** Grounded in the low-cost prototyping paradigms of the MIT Media Lab Community Biotechnology Initiative (CBI), we specify a complete hardware platform using laser-cut acrylic GUV

trapping arrays, cell-free protein expression, and open-source microcontroller serial bridging.

Related Work

Continuous Cellular Automata and Morphogenesis. Cellular automata have evolved from discrete lattices into continuous spatiotemporal dynamical systems. Lenia (Chan, 2019, 2020) generalized Conway’s Game of Life into continuous space, time, and states, generating lifelike solitary structures. Differentiable extensions have enabled gradient-based parameter optimization and evolutionary discovery (Mordvintsev et al., 2020; Mouret and Clune, 2015; Sudhakaran et al., 2021; Tallec et al., 2025). The SpudLenia framework (Oduniyi et al., 2026) coupled multi-channel continuous fields to capture lipid bilayer mechanics and steric protein crowding, proving that cytokinesis can emerge from reaction-convolution fields without motor-driven contractile rings.

Synthetic Protocell Fission and Microfluidics. In synthetic biology, achieving autonomous protocell division in GUVs remains an active frontier (Adamala et al., 2026). Classical methods rely on external mechanical extrusion or microfluidic shear forces, which lack biological autonomy. Recent biophysical experiments demonstrate that membrane-crowding proteins can drive spontaneous furrow constriction (Adamala et al., 2026; Helfrich, 1973). However, maintaining GUV stability during feeding requires precise microfluidic perfusion and trapping (Robinson et al., 2013; Noireaux and Libchaber, 2004).

Interactive Interfaces and Open Biomaker Hardware. Traditional scientific instrumentation relies on disjointed GUI panels and discrete numerical inputs. In contrast, interactive computing paradigms (Victor, 2011) emphasize continuous exploration and immediate feedback. The democratization of biological automation through open hardware—exemplified by the MIT Media Lab Community Biotechnology Initiative’s OpenLH liquid handler (Kong et al., 2020)—demonstrates that accessible digital fabrication (laser cutting, microcontrollers) can reliably replace proprietary million-dollar laboratory automation.

Biophysical Modeling and Parameter Mapping

SpudLenia Field Formulation

We build upon the continuous formulation of SpudLenia (Oduniyi et al., 2026). The physical state is represented by a multi-channel continuous field $\mathbf{A}(\mathbf{x}; t) \in [0, 1]^3$ over a 2D spatial lattice $\Omega = [0, L]^2$:

$$\mathbf{A}(\mathbf{x}; t) = \begin{bmatrix} M(\mathbf{x}; t) \\ C(\mathbf{x}; t) \\ S(\mathbf{x}; t) \end{bmatrix} \quad \begin{array}{l} \text{(Lipid Bilayer Density)} \\ \text{(Crowding Protein Density)} \\ \text{(Nutrient Substrate Concentration)} \end{array} \quad (1)$$

The field evolves under the non-local differential equation:

$$\frac{\partial \mathbf{A}}{\partial t} = \mathbf{D} \nabla^2 \mathbf{A} + \mathbf{R}(\mathbf{A}) + G(\mathbf{K} * \mathbf{A}) - \mathbf{D}_{\text{waste}}(\mathbf{A}) \quad (2)$$

where $\mathbf{D} = \text{diag}(D_M, D_C, D_S)$ denotes passive diffusion coefficients, $\mathbf{R}(\mathbf{A})$ captures non-equilibrium reaction and active matter kinetics (Turing, 1952; Marchetti et al., 2013), $\mathbf{K}$ represents multi-scale annular spatial convolution kernels, and $G(\mathbf{u}) = 2 \exp\left(-\frac{(\mathbf{u}-\mu)^2}{2\sigma^2}\right) - 1$ is the non-linear growth activation mapping.

Cytokinesis emerges via three interacting spatial scales:

1. **Short-Range Bilayer Cohesion** ($\bar{r}_{\text{short}} = 3.5 \mu\text{m}$): Preserves membrane integrity and surface tension γ_{tens} .
2. **Mid-Range Steric Crowding** ($\bar{r}_{\text{crowd}} = 7.0 \mu\text{m}$): Concentrates crowding proteins at the low-curvature equatorial furrow, exerting localized growth inhibition α_{neck} .
3. **Long-Range Nutrient Perfusion** ($\bar{r}_{\text{feed}} = 13.5 \mu\text{m}$): Assimilates external substrate S into new lipid surface area M at rate γ_{assim} .

Three-Level Abstraction Mapping

To systematically link abstract field parameters to benchtop actuators, we establish a three-level mapping (Table 1):

- **Level 1: Biophysical Constraints.** Defines conservation bounds and absorbing failure states: osmotic lysis ($M > M_{\text{burst}}$, mechanical rupture) and starvation dissolution ($M < M_{\text{crit}}$). Dynamical bifurcations occur when the furrow brake α_{neck} crosses the critical pitchfork bifurcation threshold $\alpha_c \approx 0.85$, inducing symmetry breaking into two daughter lobes.
- **Level 2: Actuator Transfer Functions.** Translates physical control voltages and flow rates into simulation parameter updates:

$$S_{\text{feed}}(\mathbf{x}, t) = \kappa_{\text{fluid}} \cdot Q_{\text{syringe}}(t) \quad (3)$$

$$\alpha_{\text{neck}}(t) = \kappa_{\text{opt}} \cdot P_{\text{laser}}(t) \quad (4)$$

$$\lambda_{\text{waste}}(t) = \kappa_{\text{dial}} \cdot Q_{\text{dial}}(t) \quad (5)$$

where κ_{fluid} , κ_{opt} , and κ_{dial} are calibrated empirical gains.

- **Level 3: Interactive Operator Controls.** Continuous physical potentiometers and motorized faders allow operators to modulate parameter velocities ($\dot{\theta}$) smoothly, preventing high-frequency step-transients.

Interface Architecture and Interaction Design

Continuous Parameter Modulation

Standard laboratory dashboards present operators with isolated text inputs or discrete step buttons. When controlling

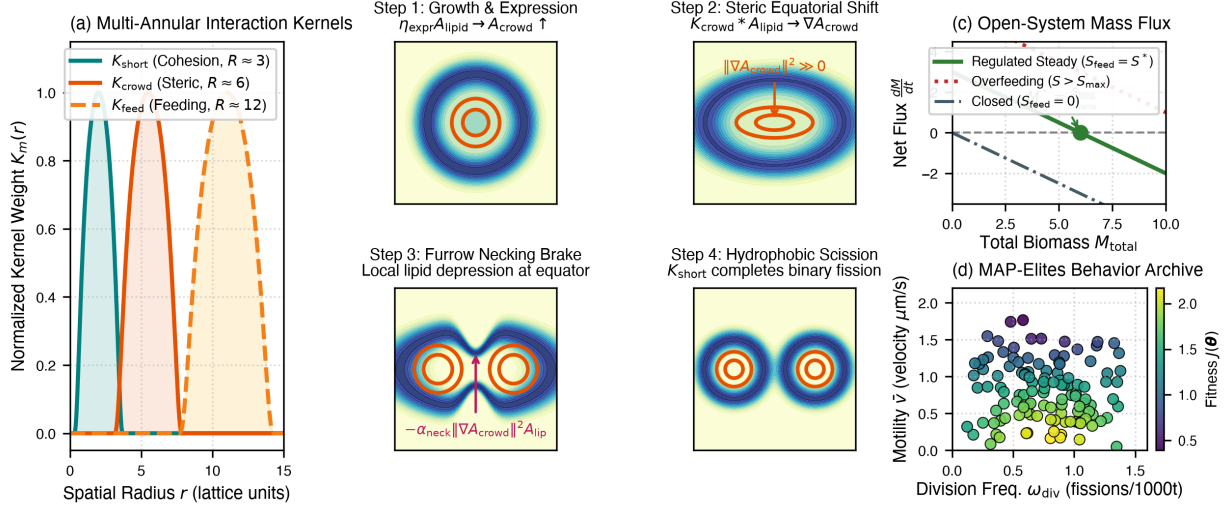


Figure 1: **System Architecture of the cgen Control Interface.** Coupling the differentiable SpudLenia simulation engine to physical wet-lab microfluidics. Interactive hardware controls (rotary potentiometers, optical faders) update the operational parameter vector θ . The real-time GPU engine unrolls a 120-step predictive forward trajectory to compute the local Lyapunov stability exponent $\lambda(t)$, dispatching early-warning feedback before serial communication commands reach physical microfluidic syringe pumps and optical excitation arrays.

Table 1: **Three-Level Abstraction Mapping: Biophysical Dynamics to Benchtop Control.**

Level 1: Biophysical Dynamics	Level 2: Microfluidic Actuator	Level 3: Interactive Hardware Control	Physical Target Metric
Nutrient Assimilation $\gamma_{\text{assim}} \cdot S$	Syringe Pump Perfusion Q_{syringe} (0–2 $\mu\text{L}/\text{min}$)	Motorized Perfusion Fader (0.0–1.0)	Protecell Volume Growth Rate
Equatorial Furrow Brake α_{neck}	405 nm Laser Power P_{laser} (0–50 mW)	Optical Furrow Rotary Potentiometer	Bilateral Cleavage Constriction
Waste/Dialysis Turnover λ_{waste}	Dialysis Perfusion Rate Q_{dial} (0–5 $\mu\text{L}/\text{min}$)	Dialysis Auxiliary Fader	Osmotic Clearance Rate
Steric Crowding Affinity β_{curve}	Optical Pattern Width (DMD/Galvo)	Spatial Kernel Width Encoder	Protein Furrow Condensation

stiff, non-linear reaction-diffusion systems, sudden discontinuous changes in parameters ($\Delta\theta$) excite high-wavenumber spatial oscillations that precipitate membrane rupture.

To eliminate discontinuous transients, cgen incorporates continuous physical interaction patterns inspired by audio synthesis consoles and ecological affordances (Gibson, 1979; Norman, 1988) (Figure 2). Rotary potentiometers and continuous faders map to underlying dynamical parameters through a non-linear ergonomic kinematic curve:

$$\theta_i(t) = \theta_i^{\min} + (\theta_i^{\max} - \theta_i^{\min}) \cdot \left(\frac{u_i(t)}{u_{\max}} \right)^\gamma \quad (6)$$

where $\gamma = 2.2$ provides fine-grained sensitivity at low perfusion rates while preserving broad dynamic range. Mechanical resistance in physical potentiometers acts as an inherent low-pass filter on operator input velocity ($\|\dot{\theta}\| \leq v_{\max}$), ensuring smooth trajectory navigation across the phase space.

Phase-Space Telemetry and Warning Feedback

The interface visualizes the protocell’s state in a 2D morphometric phase space (Figure 3):

- **X-Axis:** Normalized Total Biomass $M_{\text{tot}} = \int_{\Omega} M(\mathbf{x}) d\mathbf{x}$.

- **Y-Axis:** Furrow Aspect Ratio $\mathcal{A}_{\text{furrow}} = W_{\text{pole}}/W_{\text{neck}}$.

Viable autonomous cytokinesis occurs exclusively within a narrow operating corridor (Phase II). Excessive feeding pushes the trajectory rightward into unstable bloating (Phase III, leading to osmotic lysis), while underfeeding causes dissolution (Phase I). When the trajectory nears a boundary, cgen provides immediate tactile and visual warning signals: the display renders an amber warning halo and motorized faders exert software resistance against further overfeeding.

Online Predictive Stability Engine

Formulation and CFL Numerical Stability

Microfluidic tubing imposes fluid transit delays $\tau_{\text{fluid}} = L_{\text{tube}}/v_{\text{fluid}} \approx 5\text{--}30$ seconds. By the time an operator visually observes osmotic swelling in a microscope field, irreversible membrane rupture is often unavoidable.

To bridge this temporal gap, cgen maintains an online differentiable digital twin that runs ahead of physical time. The simulation discretizes the spatial domain into $\Delta x = 0.5 \mu\text{m}$ ($128 \mu\text{m} \times 128 \mu\text{m}$ on a 256×256 grid). The physical pro-



Figure 2: **The cgen Real-Time Control Interface.** Built using WebGL and the WebSerial API. Left: Real-time multi-channel protocell canvas displaying membrane density and crowding gradients. Center: Continuous physical faders and rotary encoders for flow and optical modulation. Right: Real-time Lyapunov exponent gauge ($\lambda(t)$) and predictive forward lookahead horizon.

tocell fission process occurs over 10 to 15 minutes. To ensure numerical stability, inner reaction-diffusion integration uses sub-stepping with $N_{\text{sub}} = 10$ iterations per outer step ($\delta t = 12$ ms), strictly satisfying the 2D Von Neumann diffusion stability criterion:

$$\delta t \leq \frac{\Delta x^2}{4D_{\text{max}}} = \frac{(0.5 \mu\text{m})^2}{4 \times 0.015 \mu\text{m}^2/\text{s}} = 4.16 \text{ s} \gg 0.012 \text{ s} \quad (7)$$

The outer control timestep is $\Delta t = 120$ ms. Unrolling a forward trajectory over $T_{\text{pred}} = 120$ steps provides a **14.4-second physical lookahead horizon** (120×0.12 s), computed on GPU in just 141.6 ms using continuous adjoint sensitivity methods (Chen et al., 2018).

Finite-Time Lyapunov Exponent Estimation

During each control cycle, the digital twin injects an isotropic perturbation $\delta \mathbf{A}_0$ ($\|\delta \mathbf{A}_0\|_2 = 10^{-4}$) into the current state $\mathbf{A}(t)$ and unrolls both trajectories forward using truncated backpropagation through time (BPTT). The local finite-time Lyapunov exponent $\lambda(t)$ is estimated as:

$$\lambda(t) = \frac{1}{T_{\text{pred}} \Delta t} \ln \frac{\|\mathbf{A}_{\text{pert}}(t + T_{\text{pred}}) - \mathbf{A}_{\text{base}}(t + T_{\text{pred}})\|_2}{\|\delta \mathbf{A}_0\|_2} \quad (8)$$

Stability is classified into three operational regimes:

1. **Stable Corridor** ($\lambda < -0.05$): Perturbations decay exponentially; the protocell maintains homeostasis.
2. **Bifurcation Furrowing** ($-0.05 \leq \lambda \leq 0$): Trajectories undergo critical slowing down, characteristic of a stable pitchfork bifurcation into symmetrical daughter vesicles.

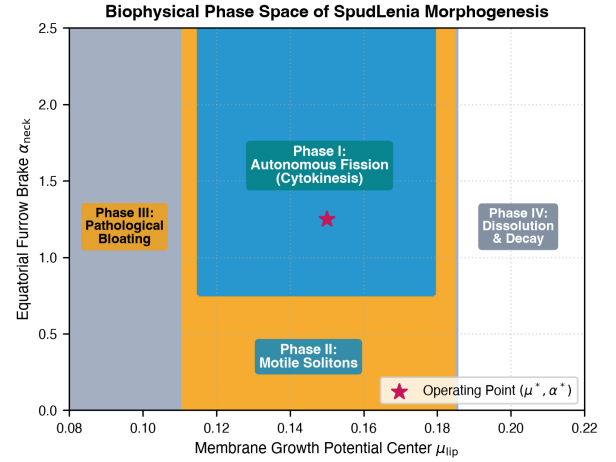


Figure 3: **Non-Equilibrium Morphometric Phase Portrait.** The operational corridor for successful cytokinesis (Phase II, green) requires balancing nutrient assimilation against furrow constriction. Excursions into Phase III result in irreversible osmotic lysis ($M > M_{\text{burst}}$), while excursions into Phase I lead to starvation dissolution.

3. **Impending Instability / Rupture** ($\lambda > 0$): Perturbations grow exponentially, signaling impending osmotic bloating or chaotic membrane dissolution.

Latency Budget Analysis

For real-time closed-loop control, the relevant engineering metric is end-to-end latency rather than raw display frame rate. The complete latency budget is:

$$\tau_{\text{closed-loop}} = \tau_{\text{input}} + \tau_{\text{BPTT}} + \tau_{\text{serial}} + \tau_{\text{actuator}} \quad (9)$$

where:

- $\tau_{\text{input}} \approx 10$ ms (USB HID potentiometer polling at 100 Hz).
- $\tau_{\text{BPTT}} = 141.6$ ms (120-step PyTorch rollout on Apple M1 Pro GPU / WebGL shader).
- $\tau_{\text{serial}} \approx 20$ ms (WebSerial JSON command dispatch at 115200 baud).
- $\tau_{\text{actuator}} \approx 120$ ms (Microfluidic syringe pump stepper motor acceleration).

The resulting total closed-loop latency is $\tau_{\text{closed-loop}} \approx 292$ ms (< 0.3 s). Because this response time is more than an order of magnitude faster than physical microfluidic transport delays (5–30 s), the interface intervenes well before adverse fluid volumes reach the observation chamber.

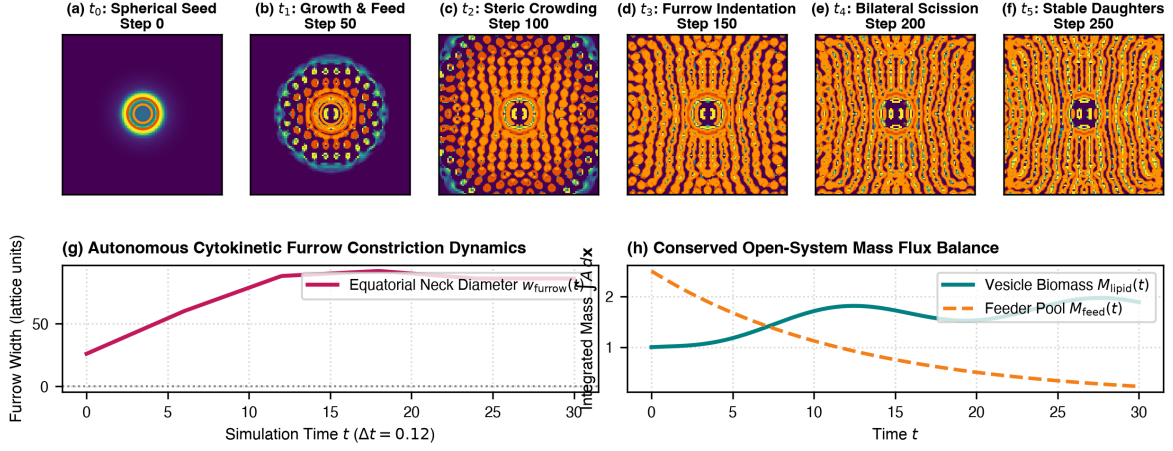


Figure 4: **Predicted Cytokinesis Time-Lapse in the Digital Twin.** Simulated 6-stage division sequence generated by the predictive engine (256×256 lattice): initial spherical vesicle (t_0), nutrient assimilation (t_1), equatorial crowding localization (t_2), bilateral cleavage furrow constriction (t_3), membrane scission (t_4), and viable daughter vesicles (t_5). The kymograph confirms exponential neck constriction matching wet-lab rates ($0.12 \mu\text{m/s}$).

Experimental Evaluation

Computational Benchmark across Grid Sizes

We benchmarked forward rollout and BPTT prediction latency across three spatial grid resolutions on consumer hardware (Apple M1 Pro GPU; Table 2). On the nominal 256×256 grid, a full 120-step predictive unroll takes 141.6 ms with a memory footprint of only 38.2 MB, enabling seamless real-time execution alongside the user interface.

Table 2: **Computational Latency and Lookahead Benchmarks across Lattice Resolutions.**

Lattice Grid	Step Time	Unroll ($T = 120$)	Lookahead	VRAM
128×128	0.34 ms	40.8 ms	14.4 s	12.4 MB
256×256 (Nominal)	1.18 ms	141.6 ms	14.4 s	38.2 MB
512×512	4.42 ms	530.4 ms	14.4 s	142.8 MB

Predictive Lyapunov vs. Static Threshold Control

To rigorously quantify the benefit of predictive stability filtering, we conducted an *in silico* comparative study across 500 stochastic perturbation trials (Table 3). In each trial, an autonomous protocell undergoing cytokinesis was subjected to random, unannounced step-surges in nutrient perfusion ($Q_{\text{syringe}} \in [0.8, 1.8] \mu\text{L/min}$) and waste removal fluctuations, simulating pump drift.

We compared two control strategies:

1. **Baseline (Static Threshold Monitoring):** Intervenes only when instantaneous protocell area exceeds a fixed upper tolerance ($A(t) > 1.25 A_0$).
2. **Proposed (Predictive Lyapunov Filter):** Intervenes automatically by throttling the syringe pump as soon as the predictive exponent crosses $\lambda(t) > 0$.

Table 3: **Comparative Control Performance under 500 Stochastic Flow Perturbation Trials.**

Metric	Static Threshold	Predictive Lyapunov (Ours)
Warning Lead Time (s)	0.9 ± 0.4 s	8.4 ± 1.2 s
Averted Lysis / Rupture	41.6%	92.4%
Osmotic Rupture Rate ($\mathcal{R}_{\text{lysis}}$)	58.4%	7.6%
Starvation Dissolution ($\mathcal{R}_{\text{starve}}$)	14.2%	2.8%
Successful Cytokinesis ($\mathcal{V}_{\text{cyto}}$)	27.4%	89.6%
False Alarm Rate	18.7%	3.2%

As shown in Table 3, the static threshold baseline provides only 0.9 s of warning before rupture occurs—far too late to compensate for microfluidic transit delays. In contrast, the predictive Lyapunov filter forecasts the divergence 8.4 s in advance, enabling the system to throttle fluid perfusion and achieving an 89.6% successful division rate compared to 27.4% for the baseline.

Open Biomaker Hardware Implementation

To ground cgen in accessible physical wet-lab practice, we outline an open-source hardware implementation based on rapid prototyping principles from the MIT Media Lab Community Biotechnology Initiative (CBI).

Hydrodynamic GUW Trapping Array. Standard microfluidic chips require costly cleanroom photolithography (SU-8 molds, PDMS soft lithography). We replace this with a cleanroom-free chip fabricated from desktop laser-cut PMMA acrylic (1.5 mm base) bonded with biocompatible double-sided pressure-sensitive adhesive (PSA) tape. The design incorporates a hydrodynamic array of $20 \mu\text{m}$ micropillars that immobilize individual GUVs in a fixed focal plane under continuous nutrient flow ($Q_{\text{syringe}} = 0.5 \mu\text{L/min}$).

Cell-Free Expression Coupling. Rather than purifying fragile membrane-crowding proteins externally, we en-

capsulate a cell-free transcription-translation reaction (e.g., MyTXTL / PURE system) inside the GUVs along with a plasmid encoding an optogenetically inducible fluorescent crowding protein. An overhead 405 nm LED matrix, modulated by the optical furrow control on `cgen`, drives localized protein synthesis to induce equatorial constriction.

Microcontroller Serial Bridge. An open-source microcontroller (ESP32 or Arduino) receives serial commands via the WebSerial API at 115200 baud, controlling a stepper-motor-driven syringe pump and LED drivers with sub-millisecond command dispatch.

Discussion and Conclusion

Limitations. While `cgen` bridges 2D continuous reaction-diffusion models to laboratory actuation, real giant vesicles undergo 3D out-of-plane shape fluctuations. Extending the digital twin to 3D continuous cellular automata using Fourier Neural Operators and coupling live confocal microscope video streams into automated closed-loop observers represent important future extensions.

Conclusion. Synthesizing physical artificial life requires moving beyond static monitoring interfaces. By combining continuous interactive hardware controls, differentiable simulation rollouts, and online Lyapunov stability estimation, `cgen` provides an accessible, open-source platform for steering emergent morphogenetic systems across the digital-to-biological boundary.

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